

Federico Palatnik Moroz

Backend Software Engineer · API Integration · Automation Systems · Distributed Systems

· Buenos Aires, Argentina · [linkedin.com/in/federico-moroz-1760a2241](https://www.linkedin.com/in/federico-moroz-1760a2241) · github.com/federicomoroz
· federicomoroz.github.io · English C2 · Remote · UTC-3, full overlap with US business hours

PROFESSIONAL SUMMARY

Backend Software Engineer building REST APIs, third-party platform integrations, asynchronous workflows, queue systems and distributed services in Python and Java. In my most recent role I cut operational processing time from approximately 10 minutes to 5 seconds across ~2,100 monthly transactions.

I integrate AI with deterministic code: guardrails, validation and caching layers keep LLM calls predictable and cost-efficient. Clean architecture, SOLID principles and real test coverage — the code is public and auditable. Full professional proficiency in English (C2), with a track record of delivering projects end to end in English. Based in Buenos Aires (UTC-3), which overlaps the entire US business day.

TECHNICAL SKILLS

Areas: Backend Development, Distributed Systems, Microservices, Queue Systems, Asynchronous Processing, Process Automation, Database Design, AI Integration, LLM Orchestration

Languages: Python, Java, JavaScript, TypeScript, C#, SQL, PHP

Frameworks & Libraries: FastAPI, Flask, Node.js, React, SQLAlchemy, Alembic, Pydantic v2, dependency-injector, httpx, Spring Boot, Spring Data JPA, Spring Kafka, Hibernate, Maven, .NET 8, ASP.NET Core, Entity Framework Core, Django, Laravel

Databases & Infrastructure: Apache Kafka, Redis, PostgreSQL, MySQL, Flyway, SQLite, Qdrant (Vector DB), Docker, Docker Compose, Gunicorn, APScheduler, Render

AI & Integration: Claude API (Anthropic), Voyage AI, Langfuse, RAG, Vector Search, Guardrails, LLM-as-Judge, Prompt Caching, Batch API, SSE Streaming, WebSockets, n8n, Airtable, BigQuery, REST APIs, Webhooks, SendGrid, Telegram Bot API

Testing & Quality: pytest, pytest-asyncio, pytest-cov, Playwright, mypy, Ruff, pre-commit, GitHub Actions, JUnit 5, xUnit, ArchUnit, Testcontainers, multi-engine contract tests, architecture tests (AST and ArchUnit)

Architecture & Practices: OOP, SOLID, Clean Architecture, Hexagonal Architecture (Ports & Adapters), MVC, Repository Pattern, Composition Root, Event-Driven, Pipeline, Circuit Breaker, Transactional Outbox, Idempotent Consumer, Backpressure, Dependency Injection, Plugin Systems, CI/CD

Development Tools: Git, GitHub, Postman, VS Code

PROFESSIONAL EXPERIENCE

Backend Developer 2022 – Present
Freelance · International clients

- Developed a training plan generation platform on the Claude API, now in daily production use by a client.
- Delivered API integrations and automation work to international clients, working entirely in English.

Backend Developer / Automation Engineer Nov 2024 – Mar 2026
Del Mundo a Tus Manos · International e-commerce and logistics

- Integrated operations with third-party APIs — Amazon, Mercado Libre and Tienda Nube (LatAm e-commerce platforms), the national tax authority API, international couriers, and a Laravel/PHP backend — over REST, webhooks and purpose-built endpoints, handling authentication, path params, secrets management and IP allowlisting.
- Cut operational processing time from approximately 10 minutes to 5 seconds across ~2,100 monthly transactions.
- Built operator-facing interfaces in Node.js, React and TypeScript on the Interface Extensions SDK, calling external endpoints client-side to avoid consuming the platform's automation-run quota.
- Diagnosed and fixed a silent failure on batches of 100+ records where requests hung with no trace in the execution log. Resolved with a 15 s AbortController, retry with exponential backoff, and non-blocking per-record error handling.

- Designed and maintained the operational data model — 11 relational tables, ~115,000 records and 48 automations in production — with global configuration centralized in a single parameters table instead of values hardcoded per automation.
- Built internal tooling: QR label generation and an AI-assisted WhatsApp inquiry routing system.

Game Developer and Programming Instructor

2021 – 2024

Freelance

- Built contract gameplay systems and modular frameworks in Unity and Unreal Engine: combat systems, AI controllers, and tooling for other developers.
- Taught C#, JavaScript, Python, C++ and game development to students across skill levels, with code reviews and project-based mentoring.

PROJECTS

CIRI Chargeback Agent

github.com/federicomoroz/ciri-api-aux-sourcecode · ciri-chargeback-agent.onrender.com/panel · federicomoroz.github.io/en/projects/agente-contracargos/

Agent that resolves chargebacks, built as a technical challenge: RAG with semantic search over Qdrant, deterministic guardrails —the code decides, the LLM explains— and n8n as an explicit orchestrator. It grades itself with an LLM judge and caches by exact key: investigating the same transaction twice drops from 113 seconds to 2. High-risk cases stop and wait for a human analyst.

FastAPI · Claude API · Qdrant · Voyage AI · n8n · Langfuse · SQLite · 1232 Tests

AI Training Plan Generator

federicomoroz.github.io/en/projects/mega-training-system/

Web platform in daily production use by a client, generating training plans in real time on the Claude API. Plugin architecture: adding a discipline never touches the core. SSE streaming, a circuit breaker around the AI calls, dependency injection and an idempotency layer.

Flask 3.0 · Claude API · SSE · SQLite · Docker · dependency-injector · Pydantic v2 · 1704 Tests

Order Outbox Service

github.com/federicomoroz/order-outbox-service · federicomoroz.github.io/en/projects/order-outbox-service/

Two Java microservices solving the dual write problem: the order and its event are written in a single Postgres commit, and a relay publishes them to Kafka with exponential backoff. If the broker is down the order still succeeds. The consumer dedupes by event_id. The hexagonal rules are verified by ArchUnit: the build fails if a layer imports one it should not.

Java 21 · Spring Boot · Apache Kafka · PostgreSQL · Flyway · ArchUnit · Testcontainers · JUnit 5 · 86 tests (72 unit + 14 integration)

Nine more services, with the source code published

federicomoroz.github.io/en/projects/

API gateway with sliding-window rate limiting and a Redis-backed circuit breaker (202 tests) · distributed task queue with scalable workers · shipping quoter with a hexagonal architecture traced end to end, and its ASP.NET Core port · multi-channel notification routing engine · webhook receiver · exchange rate monitor · environment validation CLI · turn-based combat framework in Unity.

EDUCATION AND LANGUAGES

Education: Escuela Da Vinci — Programming (2021 – 2023) · UADE — Marketing (2007 – 2010)

Languages: Spanish — Native · English — Full Professional Proficiency (C2)